

江苏省 2022 年普通高校专转本统一考试

高等数学 答案解析

一、选择题

1. 选 A

$$\text{解: } \because \lim_{x \rightarrow 0} (1-x-1) \cdot \frac{2-x}{x} = -2$$

$$\therefore \lim_{x \rightarrow 0} (1-x)^{\frac{2-x}{x}} = e^{-2}$$

要使 $f(x)$ 在 $(-1,1)$ 内连续, 只需使 $f(x)$ 在 $x=0$ 处连续

而要使 $f(x)$ 在 $x=0$ 处连续, 则应有 $\therefore \lim_{x \rightarrow 0} f(x) = f(x)$

$$\therefore f(x) = e^{-2}$$

2. 选 C

$$\text{解: } \because \lim_{x \rightarrow -1} f(x) = \lim_{x \rightarrow -1} \frac{\sin x}{x(x^2-1)} = \infty$$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{\sin x}{x(x^2-1)} = \infty$$

$\therefore x=-1, x=1$ 均为 $f(x)$ 的第二类间断点

$f(x)$ 共有 2 个第二类间断点

(注: 用课上讲的快捷判断法, 答案立即可知)

3. 选 B

$$\text{解: } \lim_{h \rightarrow 0} \frac{f(1-ah) - f(1+ah)}{h} = (-a-a)f'(1) = -2af'(1) = -2a = 1$$

$$\therefore a = -\frac{1}{2}$$

4. 选 D

解: 由原函数的定义可知 D 正确

5. 选 D

$$\text{解: } \iint_D \sqrt{x^2 + y^2} dx dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^R r^2 dr = \frac{1}{3} \pi R^3 = \pi$$

$$\therefore R = \sqrt[3]{3}$$

6. 选 C

解: 显然 C 正确

7. 选 A

$$\text{解: } A = \begin{pmatrix} -1 & -1 & 3 \\ 1 & a & 2 \\ -1 & 0 & 2 \end{pmatrix} \xrightarrow[r_1+r_3]{r_1+r_2, -r_1+r_3} \begin{pmatrix} -1 & -1 & 3 \\ 0 & a-1 & 5 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{r_2 \leftrightarrow r_3} \begin{pmatrix} -1 & -1 & 3 \\ 0 & 1 & -1 \\ 0 & a-1 & 5 \end{pmatrix}$$
$$\xrightarrow{-(a-1)r_2+r_3} \begin{pmatrix} -1 & -1 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & a+4 \end{pmatrix}$$

$$\therefore R(A) = 2, \therefore \text{应有 } a+4=0, \text{ 即 } a=-4$$

8. 选 B

$$\text{解: } M_{41} + M_{42} + M_{43} + M_{44} = 2 + 2 - 2 - 2 = 0$$

二、填空题

9. 填 0

$$\text{解: } \lim_{n \rightarrow \infty} \frac{\sin n}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sin n = 0 \quad (\text{无穷小与有界函数的积仍是无穷小})$$

10. 填 1

$$\text{解: } f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{x^2}{x \arctan x} = \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1$$

11. 填 0

$$\text{解: } f^{(2022)}(x) = 3^{2022} \cdot \sin\left(3x + \frac{2022}{2}\pi\right)$$

$$f^{(2022)}(0) = 3^{2022} \cdot \sin(3x + 1011\pi) \Big|_{x=0} = 0$$

12. 填-2

$$\begin{aligned} \text{解: } \int_a^{+\infty} 2e^{-2x} dx &= -\lim_{b \rightarrow +\infty} \int_a^b e^{-2x} d(-2x) = -\lim_{b \rightarrow +\infty} e^{-2x} \Big|_a^b \\ &= \lim_{b \rightarrow +\infty} (e^{-2a} - e^{-2b}) = e^{-2a} = e^4 \\ \therefore a &= -2 \end{aligned}$$

13. 填 $\left(\frac{1}{2}, \frac{3}{2}\right)$

$$\begin{aligned} \text{解: 令 } \lim_{n \rightarrow \infty} \left| \frac{(n+1)x^{n+1}}{a^{n+1}} \cdot \frac{a^n}{nx^n} \right| &= |x| \lim_{n \rightarrow \infty} \left| \frac{1}{a} \right| < 1 \\ |x| < \lim_{n \rightarrow \infty} |a|, \text{ 收敛半径为 } 2, \therefore \lim_{n \rightarrow \infty} |a| &= 2 \\ \text{对于 } \sum_{n=1}^{\infty} a^n (x-1)^n, \text{ 令 } \lim_{n \rightarrow \infty} \left| \frac{a^{n+1}(x-1)^{n+1}}{a^n(x-1)^n} \right| &= |x-1| \lim_{n \rightarrow \infty} |a| < 1 \\ |x-1| < \frac{1}{\lim_{n \rightarrow \infty} |a|}, \text{ 收敛半径 } R = \frac{1}{\lim_{n \rightarrow \infty} |a|} &= \frac{1}{2} \\ \sum_{n=1}^{\infty} a^n (x-1)^n \text{ 收敛域的中心点为 } x &= 1 \\ \therefore \text{收敛区间为 } \left(\frac{1}{2}, \frac{3}{2}\right) \end{aligned}$$

14. 填4

$$\begin{aligned} \text{解: } A &= \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 2 & 0 & 1 & k \\ 0 & 2 & 1 & 2 \end{pmatrix} \xrightarrow{-2r_1+r_3} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & -2 & 1 & k-4 \\ 0 & 2 & 1 & 2 \end{pmatrix} \xrightarrow{r_3+r_4} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & -2 & 1 & k-4 \\ 0 & 0 & 2 & k-2 \end{pmatrix} \\ &\xrightarrow{r_2 \leftrightarrow r_3} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & -2 & 1 & k-4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & k-2 \end{pmatrix} \xrightarrow{-2r_3+r_4} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & -2 & 1 & k-4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & k-4 \end{pmatrix} \end{aligned}$$

$\therefore$ 向量组线性相关

$\therefore$ 其秩 < 4 , 故应有 $k - 4 = 0$, $\therefore k = 4$

三、计算题

15. 解: $\lim_{x \rightarrow 0} \frac{e^{\sin x} - \sin x - 1}{x \sin x} = \lim_{x \rightarrow 0} \frac{e^{\sin x} - \sin x - 1}{x^2} = \lim_{x \rightarrow 0} \frac{e^{\sin x} \cos x - \cos x}{2x}$

$$= \lim_{x \rightarrow 0} \frac{\cos x (e^{\sin x} - 1)}{2x} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2}$$

16. 解: $\int x \arctan \frac{1}{x} dx = \frac{1}{2} \int \arctan \frac{1}{x} dx^2 = \frac{1}{2} x^2 \arctan \frac{1}{x} - \frac{1}{2} \int x^2 d \arctan \frac{1}{x}$

$$= \frac{1}{2} x^2 \arctan \frac{1}{x} - \frac{1}{2} \int x^2 \cdot \frac{1}{1 + \frac{1}{x^2}} \cdot \frac{-1}{x^2} dx = \frac{1}{2} x^2 \arctan \frac{1}{x} + \frac{1}{2} \int \frac{x^2 + 1 - 1}{1 + x^2} dx$$
$$= \frac{1}{2} x^2 \arctan \frac{1}{x} + \frac{1}{2} \int dx - \frac{1}{2} \int \frac{1}{1 + x^2} dx$$
$$= \frac{1}{2} x^2 \arctan \frac{1}{x} + \frac{1}{2} x - \frac{1}{2} \arctan x + c$$

17. 解: $\int_{-1}^5 f(x) dx = \int_{-1}^1 \frac{x^3}{\sqrt{1+x^2}} dx + \int_1^5 \frac{\sqrt{2x-1}}{\sqrt{2x-1}+1} dx = 0 + \int_1^5 \frac{\sqrt{2x-1}}{\sqrt{2x-1}+1} dx$

对于 $\int_1^5 \frac{\sqrt{2x-1}}{\sqrt{2x-1}+1} dx$, 令 $\sqrt{2x-1} = t$, 则 $x = \frac{t^2+1}{2}$

当 $x=1$ 时 $t=1$, 当 $x=5$ 时 $t=3$

$$\therefore \int_1^5 \frac{\sqrt{2x-1}}{\sqrt{2x-1}+1} dx = \int_1^3 \frac{t}{t+1} d \frac{t^2+1}{2} = \int_1^3 \frac{t^2-1+1}{t+1} dt$$
$$= \int_1^3 (t-1) dt + \int_1^3 \frac{1}{1+t} dt = \frac{1}{2} (t-1)^2 \Big|_1^3 + \ln|t+1| \Big|_1^3 = 2 + \ln 2$$

18. 解: 方程两边对 x 求偏导数, 得:

$$e^z \cdot \frac{\partial z}{\partial x} = y + y \cdot \frac{\partial z}{\partial x} + z + x \cdot \frac{\partial z}{\partial x}, \therefore \frac{\partial z}{\partial x} = \frac{y+z}{e^z - x - y}$$

方程两边对 y 求偏导数, 得:

$$e^z \cdot \frac{\partial z}{\partial y} = x + z + y \cdot \frac{\partial z}{\partial y} + x \cdot \frac{\partial z}{\partial y}, \therefore \frac{\partial z}{\partial y} = \frac{x+z}{e^z - x - y}$$

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = \frac{y+z}{e^z - x - y} dx + \frac{x+z}{e^z - x - y} dy$$

19. 解: 对应齐次方程的特征方程 $r^2 + 2r - 3 = 0$, 特征根 $r_1 = 1, r_2 = -3$

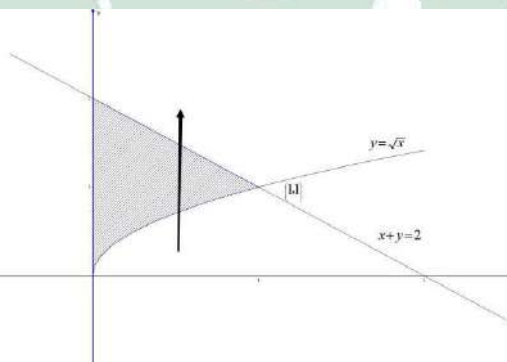
$$\text{其通解 } \bar{y} = c_1 e^x + c_2 e^{-3x}$$

$$\text{设原方程的一个特解 } y^* = A x e^x$$

$$\text{将 } y^* \text{ 代入微分方程, 解得 } A = 1, \therefore y^* = x e^x$$

$$\text{原方程的通解 } y = \bar{y} + y^* = c_1 e^x + c_2 e^{-3x} + x e^x$$

$$\begin{aligned} 20. \text{ 解: } \iint_D xy dx dy &= \int_0^1 dx \int_{\sqrt{x}}^{2-x} xy dy = \frac{1}{2} \int_0^1 dx \cdot xy^2 \Big|_{\sqrt{x}}^{2-x} = \frac{1}{2} \int_0^1 x[(2-x)^2 - x] dx \\ &= \frac{1}{2} \int_0^1 (4x - 5x^2 + x^3) dx = \frac{7}{24} \end{aligned}$$



21. 解: $AX = B$ 两边左乘 A^{-1} , 得 $X = A^{-1}B$

$$(A:E) = \begin{pmatrix} -1 & 1 & 0 & 1 & 0 & 0 \\ -1 & 2 & 0 & 0 & 1 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{-r_1} \begin{pmatrix} 1 & -1 & 0 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 & 1 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{\substack{r_1+r_2 \\ -r_1+r_3}} \begin{pmatrix} 1 & -1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 2 & -1 & 1 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{r_2+r_1 \\ -2r_2+r_3}} \begin{pmatrix} 1 & 0 & 0 & -2 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 3 & -2 & 1 \end{pmatrix}$$

$$\xrightarrow{-r_3} \begin{pmatrix} 1 & 0 & 0 & -2 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -3 & 2 & -1 \end{pmatrix}$$

$$\therefore A^{-1} = \begin{pmatrix} -2 & 1 & 0 \\ -1 & 1 & 0 \\ -3 & 2 & -1 \end{pmatrix}$$

$$X = A^{-1}B = \begin{pmatrix} -2 & 1 & 0 \\ -1 & 1 & 0 \\ -3 & 2 & -1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 5 & 6 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

22. 解: $\bar{A} = \begin{pmatrix} 1 & 1 & 1 & -3 & 6 \\ 1 & -5 & 1 & 3 & -6 \\ 1 & -2 & 2 & -1 & 2 \end{pmatrix} \xrightarrow{\substack{-r_1+r_2 \\ -r_1+r_3}} \begin{pmatrix} 1 & 1 & 1 & -3 & 6 \\ 0 & -6 & 0 & 6 & -12 \\ 0 & -3 & 1 & 2 & -4 \end{pmatrix}$

$$\xrightarrow{\substack{-\frac{1}{6}r_2 \\ 3r_2+r_3}} \begin{pmatrix} 1 & 1 & 1 & -3 & 6 \\ 0 & 1 & 0 & -1 & 2 \\ 0 & -3 & 1 & 2 & -4 \end{pmatrix} \xrightarrow{\substack{-r_2+r_1 \\ 3r_2+r_3}} \begin{pmatrix} 1 & 0 & 1 & -2 & 4 \\ 0 & 1 & 0 & -1 & 2 \\ 0 & 0 & 1 & -1 & 2 \end{pmatrix}$$

$$\xrightarrow{-r_3+r_1} \begin{pmatrix} 1 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & -1 & 2 \\ 0 & 0 & 1 & -1 & 2 \end{pmatrix}$$

x_1, x_2, x_3 为约束变量, x_4 为自由变量, 一般解 $\begin{cases} x_1 = 2 + x_4 \\ x_2 = 2 + x_4 \\ x_3 = 2 + x_4 \end{cases}$

令 $x_4 = 0$ 得原方程的一个特解 $\eta^* = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 0 \end{pmatrix}$

导出组的一般解 $\begin{cases} x_1 = x_4 \\ x_2 = x_4 \\ x_3 = x_4 \end{cases}$, 令 $x_4 = 1$ 得基础解系 $\xi = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$

原方程组通解 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 0 \end{pmatrix} + k \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad (k \text{ 为任意常数})$

四、证明题

23. 证明: 令 $F(x) = \int_0^x e^t \cos t dt - x$, 则 $F(0) = 0$

$$F'(x) = e^x \cos x - 1, \quad F'(0) = 0$$

$$F''(x) = e^x \cos x - e^x \sin x = e^x (\cos x - \sin x)$$

当 $0 < x < \frac{\pi}{4}$ 时, $F''(x) > 0$, $\therefore F'(x)$ 单调递增, $F'(x) > F'(0) = 0$

由 $F'(x) > 0$ 又可知 $F(x)$ 单调递增, 从而 $F(x) > F(0) = 0$

$$\text{即 } \int_0^x e^t \cos t dt - x > 0, \quad \therefore \int_0^x e^t \cos t dt > x$$

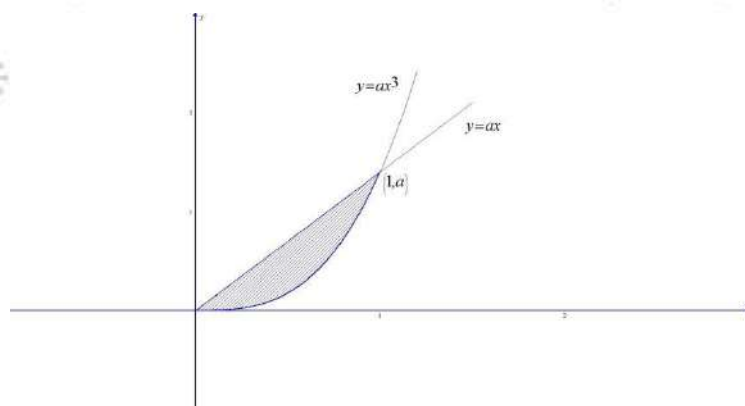
五、综合题

24. 解: $V_x = \int_0^1 \pi(ax)^2 dx - \int_0^1 \pi(ax^3)^2 dx = \frac{4\pi}{21} a^2$

$$\begin{aligned} V_y &= \int_0^a \pi \left(\sqrt[3]{\frac{y}{a}} \right)^2 dy - \int_0^a \pi \left(\frac{y}{a} \right)^2 dy = \frac{1}{\sqrt[3]{a^2}} \pi \int_0^a y^{\frac{2}{3}} dy - \frac{1}{a^2} \pi \int_0^a y^2 dy \\ &= \frac{1}{a^{\frac{2}{3}}} \pi \cdot \frac{3}{5} y^{\frac{5}{3}} \Big|_0^a - \frac{1}{a^2} \pi \cdot \frac{1}{3} y^3 \Big|_0^a = \frac{1}{a^{\frac{2}{3}}} \pi \cdot \frac{3}{5} a^{\frac{5}{3}} - \frac{1}{a^2} \pi \cdot \frac{1}{3} a^3 \\ &= \frac{3}{5} a \pi - \frac{1}{3} a \pi = \frac{4\pi}{15} a \end{aligned}$$

由 $V_x = V_y$ 得 $\frac{4\pi}{21} a^2 = \frac{4\pi}{15} a$, $\therefore a = \frac{7}{5}$

$$S_D = \int_0^1 \left(\frac{7}{5} x - \frac{7}{5} x^3 \right) dx = \left(\frac{7}{10} x^2 - \frac{7}{20} x^4 \right) \Big|_0^1 = \frac{7}{20}$$



25. 解: (1) 由 $xf'(x) - 4f(x) = 2x^3$ 得 $f'(x) - \frac{4}{x}f(x) = 2x^2$, 这是一个一阶线性方程

$$\text{通解 } f(x) = e^{\int \frac{4}{x} dx} \left[\int 2x^2 e^{-\int \frac{4}{x} dx} dx + c \right] = e^{4\ln x} \left[\int 2x^2 e^{-4\ln x} dx + c \right]$$

$$= x^4 \left(2 \int \frac{1}{x^2} dx + c \right) = x^4 \left(-\frac{2}{x} + c \right) = cx^4 - 2x^3$$

将 $f(2) = 0$ 代入通解中, 得 $c = 1$

$$\therefore f(x) = x^4 - 2x^3$$

(2) $f(x)$ 的定义域 $(-\infty, +\infty)$

$$f'(x) = 4x^3 - 6x^2, \quad f''(x) = 12x^2 - 12x = 12x(x-1)$$

令 $f''(x) = 0$, 得 $x = 0, x = 1$

x	$(-\infty, 0)$	0	$(0, 1)$	1	$(1, +\infty)$
$f''(x)$	+	0	-	0	+
$y = f(x)$	∪	拐点(0,0)	∩	拐点(1,-1)	∪

由表可知, 曲线的凹区间为 $(-\infty, 0)$, $(1, +\infty)$, 凸区间为 $(0, 1)$

拐点为 $(0, 0)$, $(1, -1)$

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